

**Large scale
production**



- After successful production of Hybridoma cells, number of cells is small, for large factories production of Mab doesn't rely on hundreds of Cells only, these factory are produce large Quantities of Mab grams / Month.

- As we said, after successful production of desired Hybridoma cells, few numbers of cells exist, cells are Propagated to increase the number of cells.

Propagation in vivo or ascitic production

- We inject the positive selected (desired) Hybridomas into the mouse.
- Inside the mouse optimal conditions (growth factors, optimal Temperature, optimal PH) this will support and Enhance the Hybridomas divide .

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- Inject 10^6 – 10^7 hybridoma cells why?
 - Injected larger than could accelerate the formation of ascitic fluid but, Hybridoma cells are cancer cells, which mean the mouse was injected with tumor cells, this could lead to Mouse death before the Ascitic fluid harvested.

- **Inject the Hybridomas IP :**

- Pristane-treated for 10-60 days before cell injection
- pristane injected IP into adult mice
- Pristane is branched chain alkane which are known to induce myeloma when injected IP.

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- Pristane is immune suppresser
 - Using Old mice (12-24 weeks)
 - Immune system is weak
 - Hybridoma are allowed to grow for 2 weeks.

- Animals must be compatible with both parent cell lines (Balb/c).
Why?

- To prevent the rejection, we don't suppresser the immune system completely.

- Within 2 weeks of cell injection, ascitic fluid 2-5 ml containing 5-15 mg/ml of Mab accumulate in the peritoneum that can be withdrawn with syringe.
- Occasionally, solid tumors will develop instead of ascites.
- This can be overcome by incubating tumor cells in vitro for a few days and then re-injection them into pristane-treated animals.

Production in animals (In-vivo)

Mouse ascites method



- Hybridoma cells injected in mouse
- Produce ascites
- Fluid contains high concentration of Ab's
- No further concentration required
- Purification required
- Easy and inexpensive
- Animal mortality

Propagation In vitro

- Allow the cells divide into cultures, large flasks.
- On the lab scale, cells can be grown in stationary culture to produce 5-50 ug/ml
- Medium scale: Monoclonal Antibodies concentration Of 30 mg/L /day have been achieved in spinner flasks.



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- Spinner flasks (stirrer bottles) : specifically designed for suspension cell culture allow for superior gas exchange and permit higher volumes of cells to be cultured. Providing a homogeneous culture environment
 - pinner flasks need semi continuous feeding
 - large scale production : by Hollow fiber bioreactor
 - Mammalian cells: have more complex nutritional needs and tolerate much lower densities than microbial cultures.

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- An increasing popular method of producing high yield of Mab in vitro is to grow hybridoma cells within a hollow fiber reactor.
 - The basic principle is that of: perfusion of culture medium with necessary CO₂ and temperature control, through a disposable hollow fiber cartridge where the cells are growing.
 - The cartridge is made up of hundreds of hollow fibers of sufficient pore size to allow medium and essential nutrients to pass through the fibers to the extracapillary space containing the growing cells,

Cartridge Hollow fiber

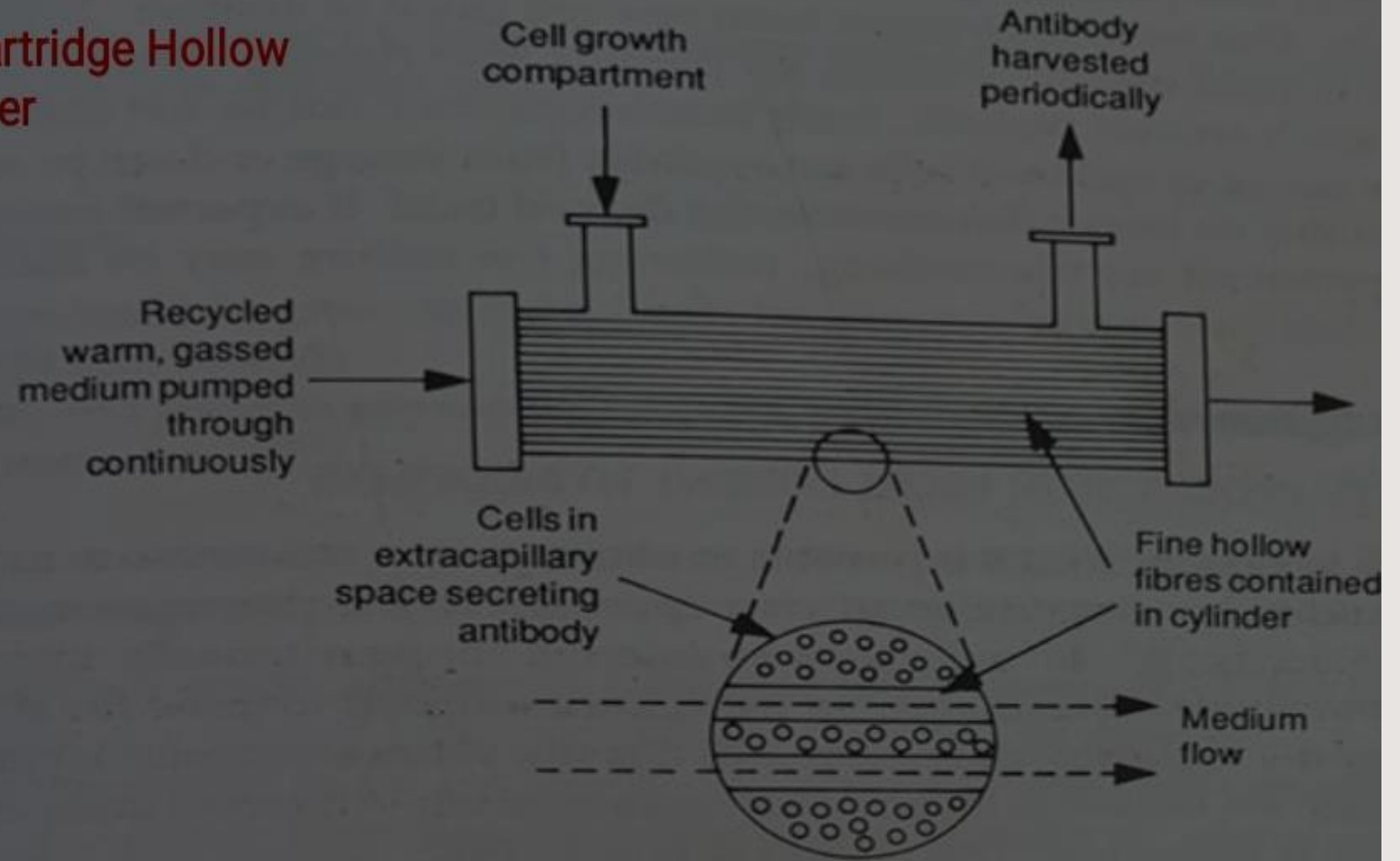
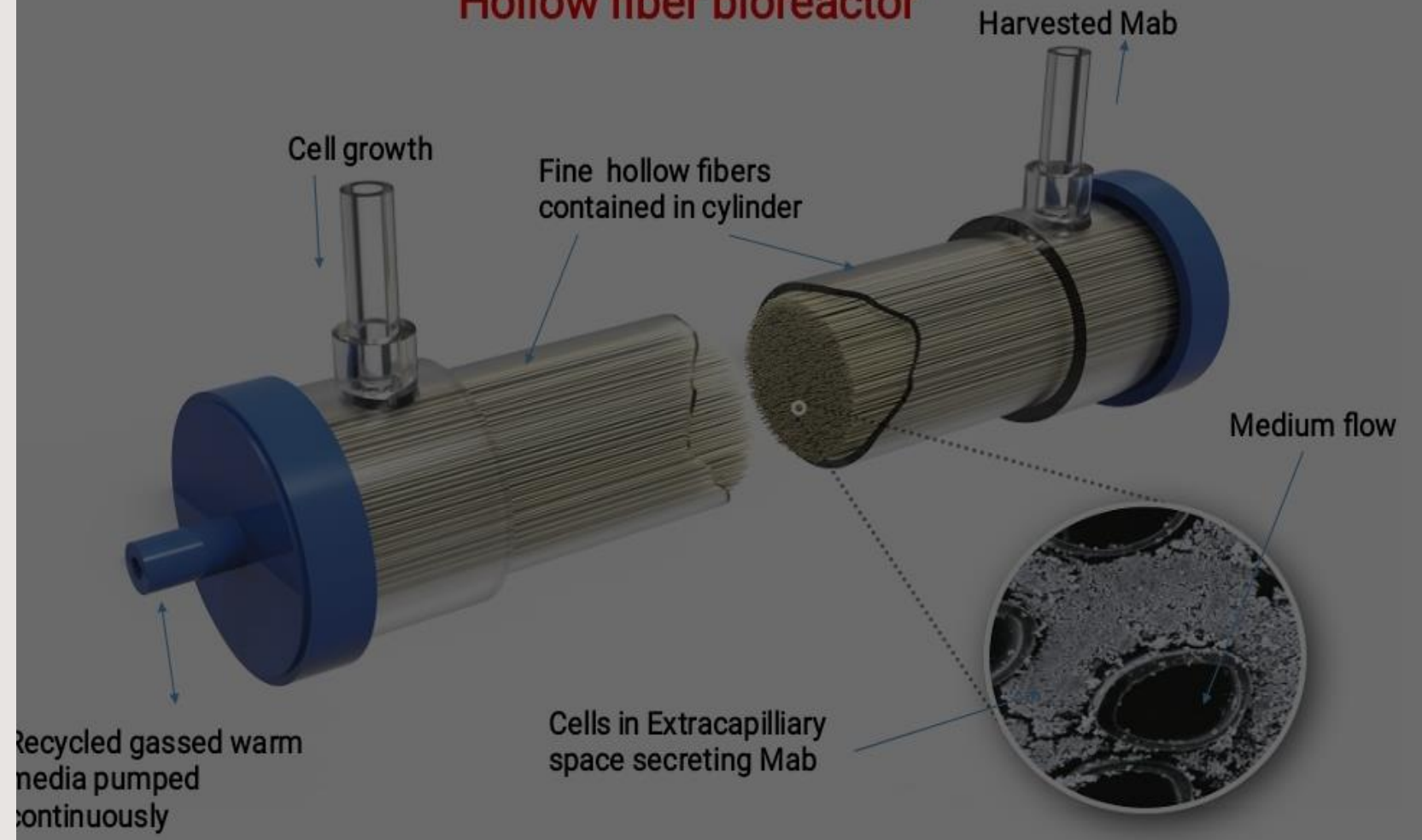
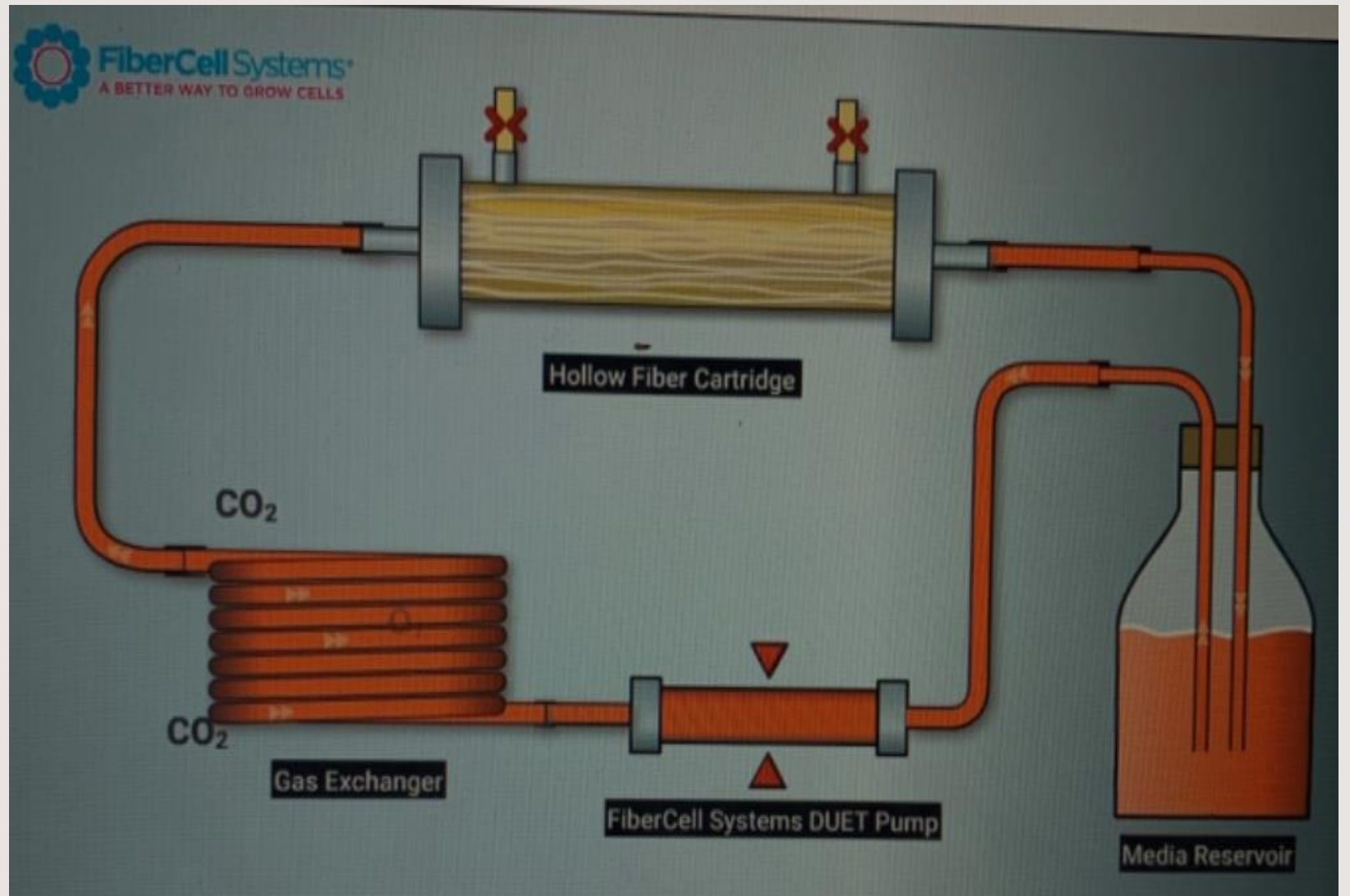


Fig. 6.3.1b Schematic diagram of hollow fibre reactor

Hollow fiber bioreactor







Hollow fiber consist of :

1.Extra capillaries space : where the Hybridoma cells present

2.Hollow fibers :where the media present

- Secreted Monoclonal Antibodies cannot pass through the fiber, are harvested periodically from the extra capillary compartment over the period of months.
- It is essential that the fidelity of the specific Mab productions be monitored regularly due to the risk that non-producing revert cells may outgrow the specific Mab-producing cells.

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- Check cells from time to time to make sure the cells are still produce Mabs.
 - Such instability is not too serious if fresh samples of the clones are available from storage and can be substituted.
 - Hybridoma cells are cancer cells, can divide indefinitely, Actually the hybridoma could be undergo spontaneous genetic mutations which lead to high risk to loss the ability to produce and secret cells.
 - This could be solved when old cells substituted with fresh cells that were storage .

Criteria of Hollow fiber Bioreactor

1.High yield with high titer product harvesting

- Enables large amounts of Mabs to be harvested periodically at high concentrations from cells maintained in continuous high-density culture.
- The fiber cell System, comprising a pump and cartridges, fits on the shelf of a standard CO₂ incubator and within only a few weeks can generate grams of Mab.
- The amount depending on the time period, clone and cartridge type used. Up to 100X greater product concentration than with flask culture is routinely possible.



2. Practical method avoids passaging cells

- A standard 20 ml Fiber Cell bioreactor can continuously maintain $> 10^9$ cells at high density over long periods in a standard incubator.
- The media bottle and cartridge module are in a closed loop that can be readily unclipped from the pump for maintenance in the laminar flow cabinet.
- Cells bound to a porous support and Cell mass is controlled during harvesting which means that passaging cells is not required.

3. Elimination of inhibitory TGF- β from hybridomas

- Preventing hybridomas from inhibiting their own productivity
- It is often overlooked that hybridomas inhibit their own growth due to the secretion of TGF- β which originates from the myeloma cell component.
- TGF- β is inhibitory cytokine
- TGF- β molecular weight = 27kd

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- therefore is small enough to diffuse across the fibres away from the cells to be diluted in the recirculating media which is periodically replaced with fresh media.

4.High Antibody avidity

- Offers a way to grow cells continuously at densities approaching in-vivo density with very constant pH and continuous elimination of waste metabolites.
- It's a reasonable assumption that this mode of culture favors natural enzymatic processing steps leading to native-like posttranslational modifications such as glycosylation that can be critical for high Ab avidity.

5. Simple disposable cartridge module with Small size

- Larger incubators can house up to 6 or more pumps operating simultaneously with higher capacity cartridge types to generate as much as 2 grams per week of a Mab.





6. Chemically Defined Medium For High Density Cell Culture (CDM-HD)

- A defined medium (also known as chemically defined medium or synthetic medium)
- is a medium in which all the chemicals used are known, no yeast, animal, or plant tissue is present.
- CDM-HD is designed specifically for the culture of cells at high density and optimized for use in the hollow fiber bioreactor systems.

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- Secreted Mabs are free of contaminating proteins from the medium and can be purified using simplified protocols, increasing net yield in many cases.
 - CDM-HD provides lot-to-lot consistency and is an economical replacement for serum.
 - It is available as a dry powder to make up one liter and is used at a concentration of 10%.
 - The cost of CDM-HD is significantly lower than serum giving you all the benefits of a chemically defined, protein-free medium without the high price.



Common Strategies for Virus Clearance in Mab Purification

- Achieving zero risk of viral contamination in biologic production is impossible. But a strategy to ensure safety of biopharmaceuticals through risk mitigation involves:

1. Confirmed absence of contaminating infectious viruses in intermediates at relevant steps in a production process.

2. Assessment of how well a manufacturing process inactivates and removes viruses.

3. Quantitative polymerase chain reaction (qPCR) can be used to assess virus removal.

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- Common Strategies for Virus Clearance in Mab

Purification Steps

1. Precipitation by ammonium sulfate 40-50%
2. Dialysis
3. Gel filtration separation according to size
4. Protein A/G sephadex affinity chromatography
5. Heat treatment
6. Viral reduction treatment
7. UV

Ammonium sulfate(AS)

- Precipitation is a method used to purify Abs by altering their solubility from ascites, serum or hybridoma supernatant. It is a specific case of a more general technique known as salting out.
- As the salt concentration (ionic strength) is increased, the solubility of the protein begins to decrease. At sufficiently high ionic strength, the protein will be almost completely precipitated from the solution (salting out).
- AS is useful for polyclonal and monoclonal IgG isolation.

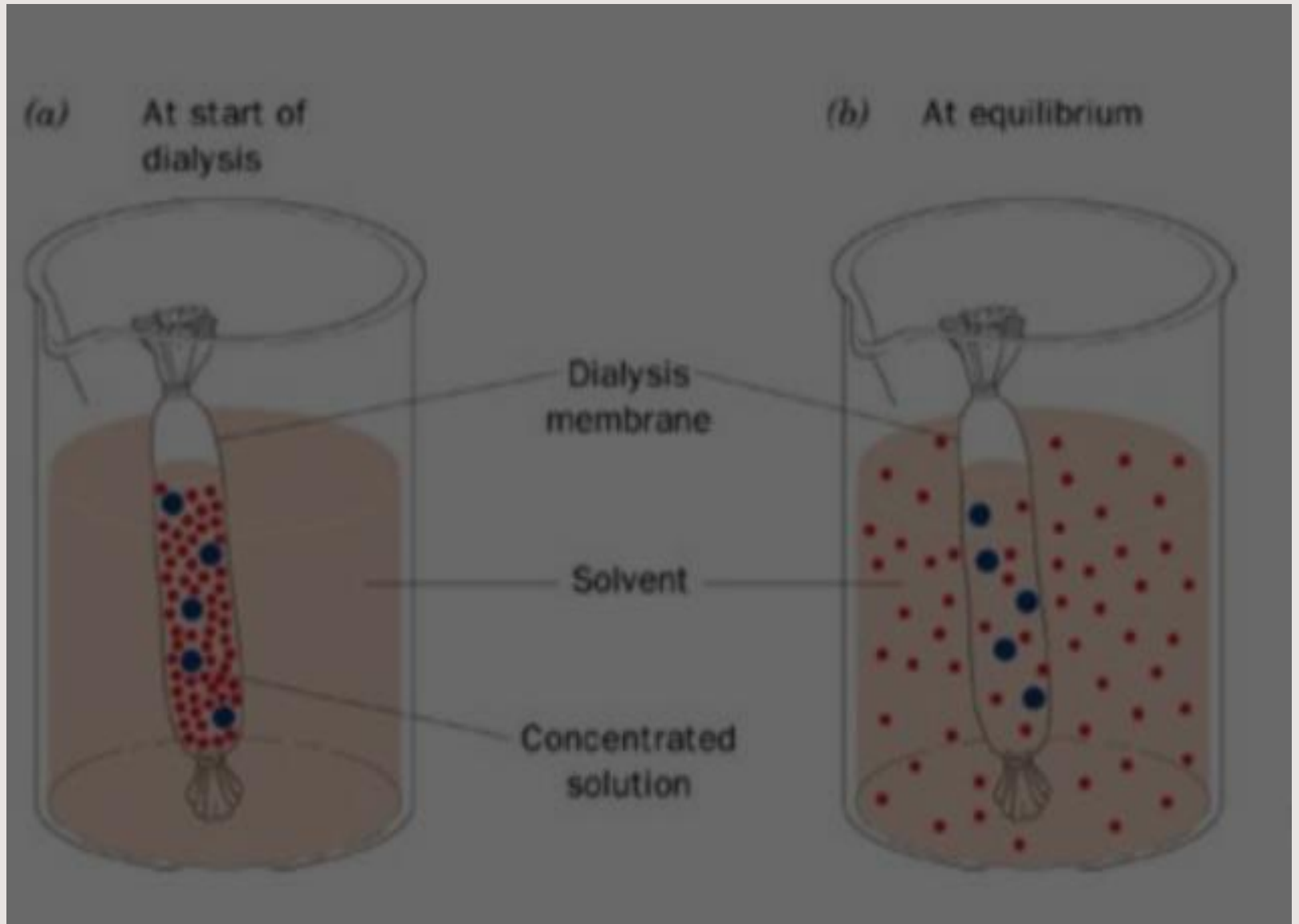
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- A solution with 35% saturated will produce a pure IgG precipitation.
 - The precipitate is collected by centrifugation, solubilized in an appropriate buffer for storage at -80 C° or dialyzed to remove residual salt prior to further purification.
 - Although fractionation with AS provides a convenient method , it should not be used as a single step purification , since the precipitated proteins contains considerable amount of contaminating proteins .

Dialysis

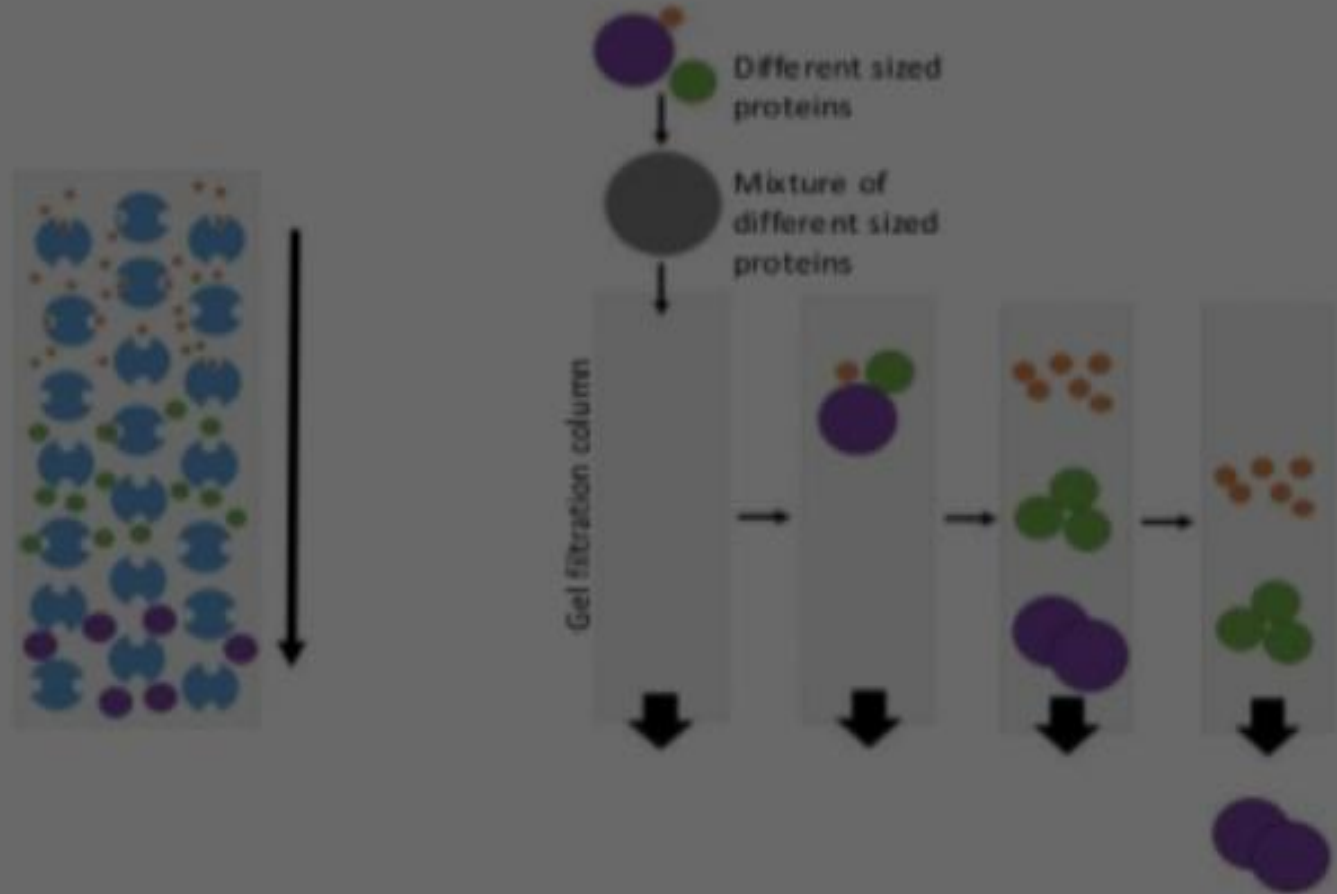
- A bag contain pores with known size (smaller than the protein of interest (Mab)
- AS can be removed by dialysis of protein solution against large volumes of desired buffer.
- The solution to be dialyzed is placed in a sealed dialysis membrane and immersed in a selected buffer.

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- Putting the solution which contain Mab and contaminates in Dialysis bag, and Immersion in Known buffer, inside Dialysis High solute concentration compare to out side, so the solutes start move from high Solute concentration (inside Dialysis) to low solute concentration (out side Dialysis) until to concentration of solutes inside Dialysis equal the concentration of solutes in surrounding buffer.

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- In addition, separation of smaller MW contaminant molecules than Igs is achieved by dialysis.
 - Dialysis membranes are available in a number of thicknesses and pore sizes.
 - Pore size is defined by “molecular weight cut-off”
 - It is necessary only to use a membrane with a pore size that is much smaller than the macromolecule of interest.



Size exclusion chromatography



Affinity Chromatography

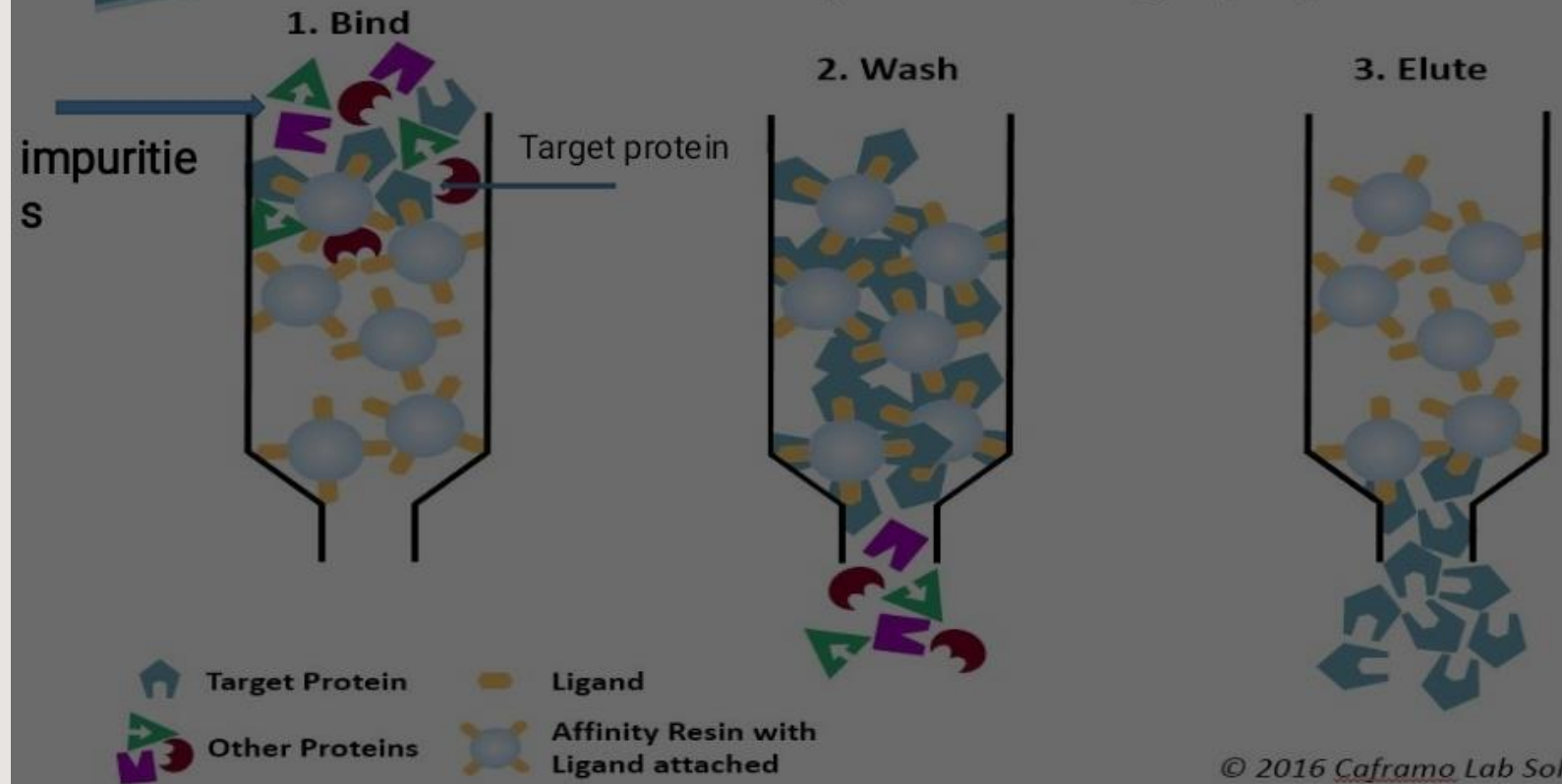
- Many proteins were founded in Bacteria have Greater Affinity to Immunoglobulins.
- Protein G in Bacteria has high affinity with IgG
- This protein has exploited in Mab purification :
- Protein G linked to Sepharose gel beads have a high capacity for binding IgG and permits

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- However, Protein G cannot discriminate between the Mab and other irrelevant IgG.
 - Which mean if the sample contain Three Types of Monoclonal Antibodies, each recognize one epitope different from others, three Monoclonal Antibodies bind to protein G.
 - bovine IgG was removed from fetal calf serum by passage over protein G before being used as part of standard tissue culture medium.

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- Mab capture is widely performed by Protein A affinity chromatography.
 - Besides the efficient removal of process-related impurities like host cell proteins and DNA, Protein A can be used for virus removal. Most viruses flow through the column.
 - As Protein A is eluted at acidic pH, a chemical inactivation of enveloped viruses by denaturation of the envelope proteins is typically performed.
 - Generally this inactivation step is performed at pH 3.0-3.4

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- As Mabs are typically basic proteins, at pH 6-8 viruses are retained by the electropositive ligand whereas most Mabs pass the column.
 - Protein G has better binding capacity for a broader range of mouse and human IgG subclasses.
 - Protein A/G is a recombinant fusion protein that includes the IgG-binding domains of both Protein A and Protein G. Protein A/G chromatography Were very effective for eliminating both enveloped and nonenveloped viruses

Protein Affinity Chromatography



Solvent/Detergent

- Treatment of plasma-derived products exist in solution with 0.3% tri(n-butyl) phosphate (TNBP)/0.2% sodium cholate (CA) can provide effective inactivation of infectious viruses.
- A detergent solubilizes a viral envelope's lipid membrane structure.
- That disruption prevents a virus from binding to or infecting cells, thus rendering it inactive.
- It is important to note that this process does not inactivate nonenveloped viruses.

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- A solvent/detergent step based on 1% triton X-100 % rapidly inactivating a range of enveloped viruses within 1 min of a 60 min treatment time.
 - Caprylic acid (CA) is described as a standard precipitation method for the isolation of IgGs from plasma. CA-induced impurity precipitation is reported as an alternative method to polishing column chromatography to intensify recombinant mAb purification processes.
 - Besides product purification, the undissociated and lipophilic form of CA acts as a virus inactivating agent by disrupting the integrity of the lipid coat of enveloped viruses.

Heat treatment

- Heat Treatment for viral inactivation usually performed above 70 °C
- This method typically causes three-dimensional changes in the structure of viral proteins, rendering viruses nonfunctional or inactive.
- Typical incubation lasts for 10 hours and is performed at 60°C
- IgG denaturation becomes significantly irreversible at temperatures higher than 65 °C, IgG almost completely loses its antigen-binding activity after heat treatment for several minutes at 90 °C.

Virus Reduction Filtration

- Many purification processes for biopharmaceuticals use virusreduction filtration as an integral part of an overall strategy for viral clearance.
- Virus-reduction filters can provide robust and effective removal of large- and medium-sized viruses.
- Such filters also can effectively remove very small viruses (e.g., parvoviruses) with pore sizes ≤ 20 nm.

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- Virus removal by virus filtration step was also confirmed to be effective for those viruses of about 50nm or greater

Ultraviolet-C (UV-C)

- Uses UV-C light low-dose radiation at 254 nm to destroy viral nucleic acids while maintaining structural and functional integrity of manufactured proteins.
- UV-C light was shown to induce very little damage to Mabs at lower fluencies and appears to be a viable option for viral inactivation in biotechnology applications.